

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

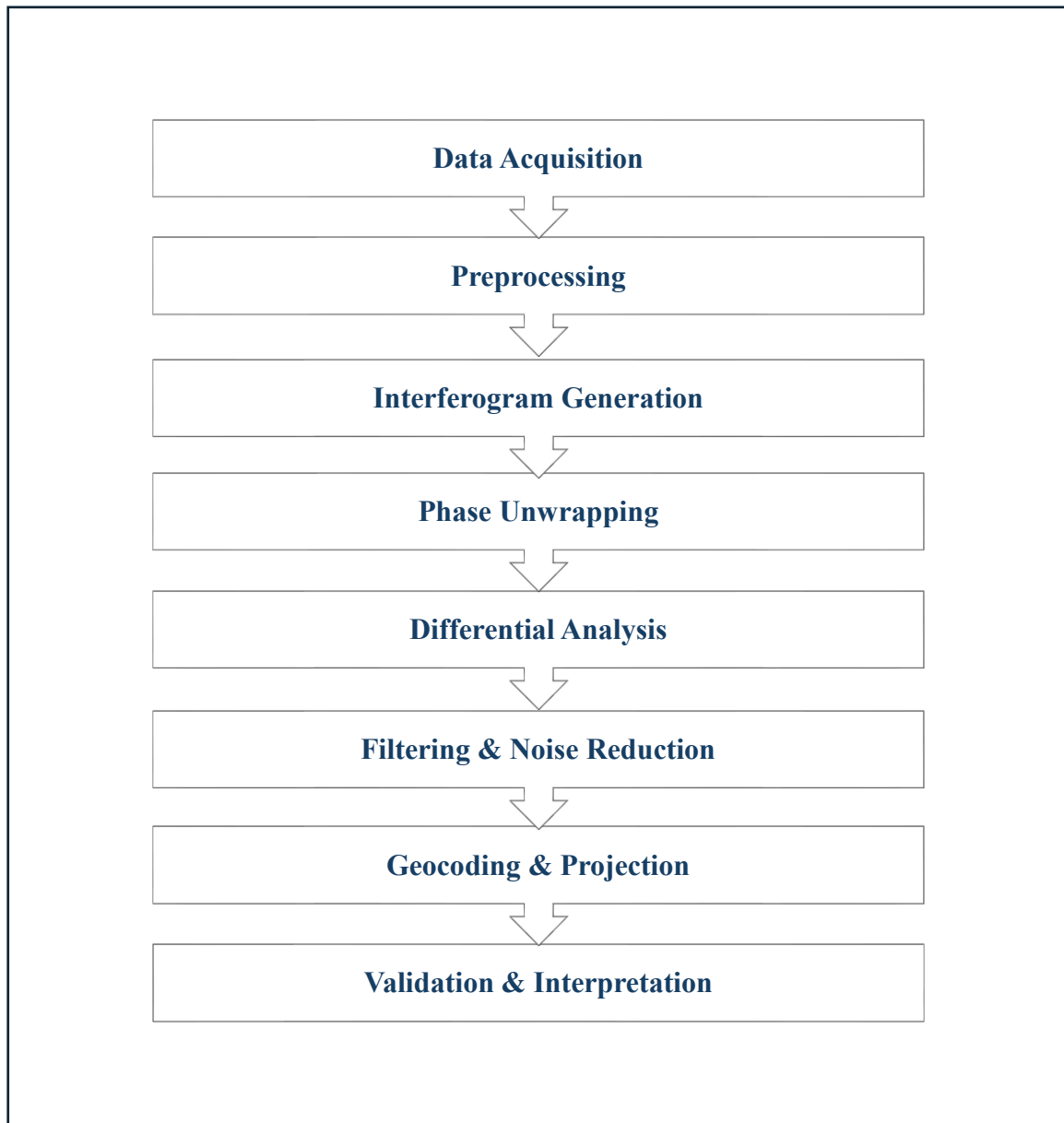
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 983+400

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

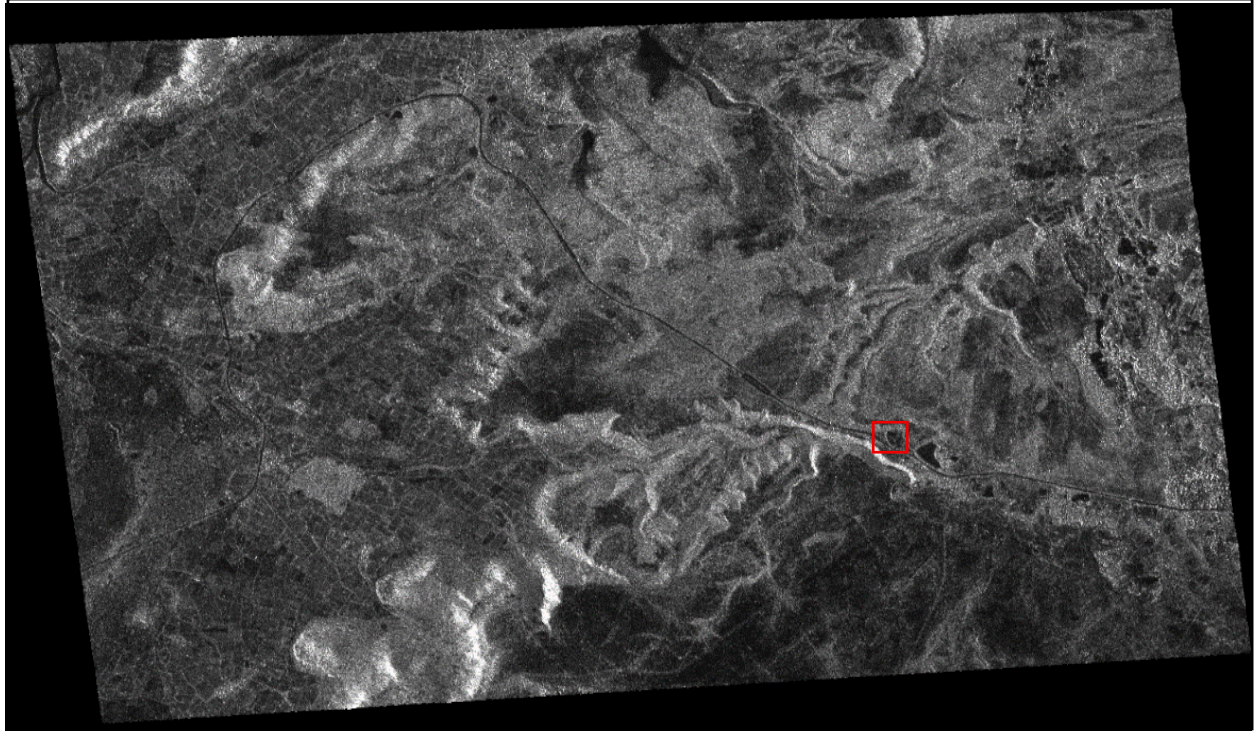
Flight Direction - DESCENDING

Polarization – VV+VH

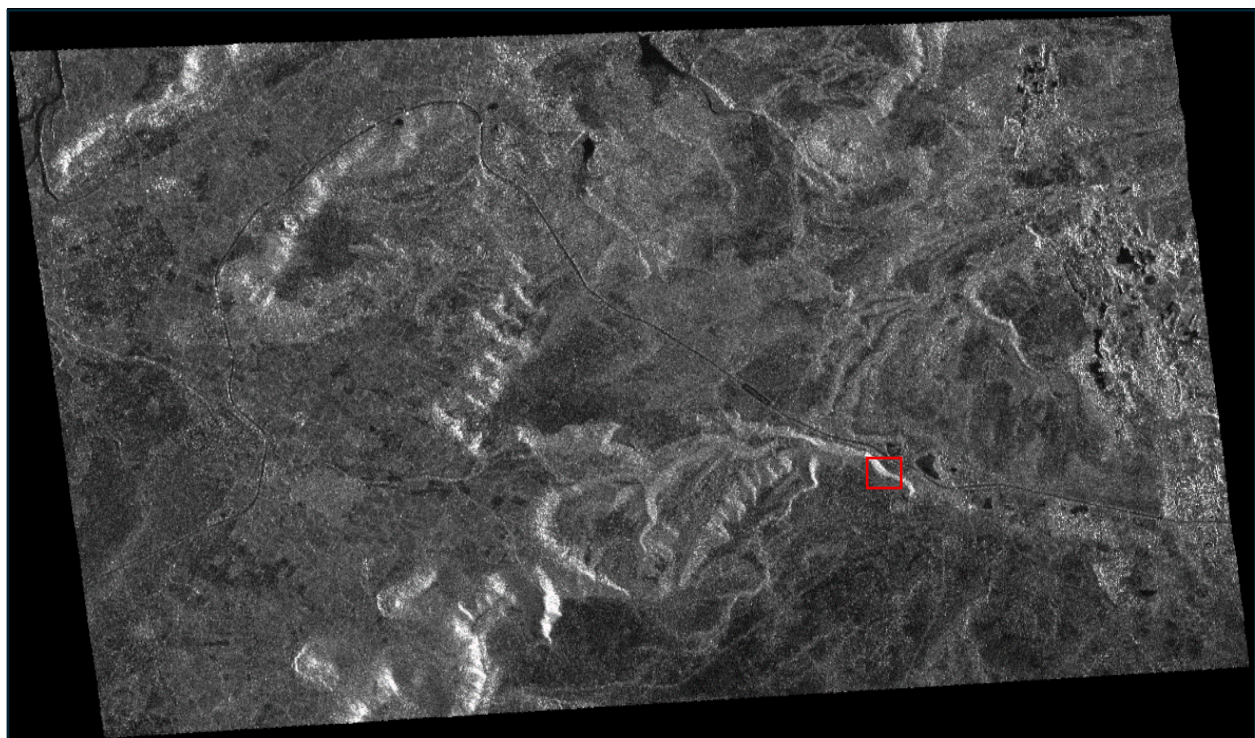
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

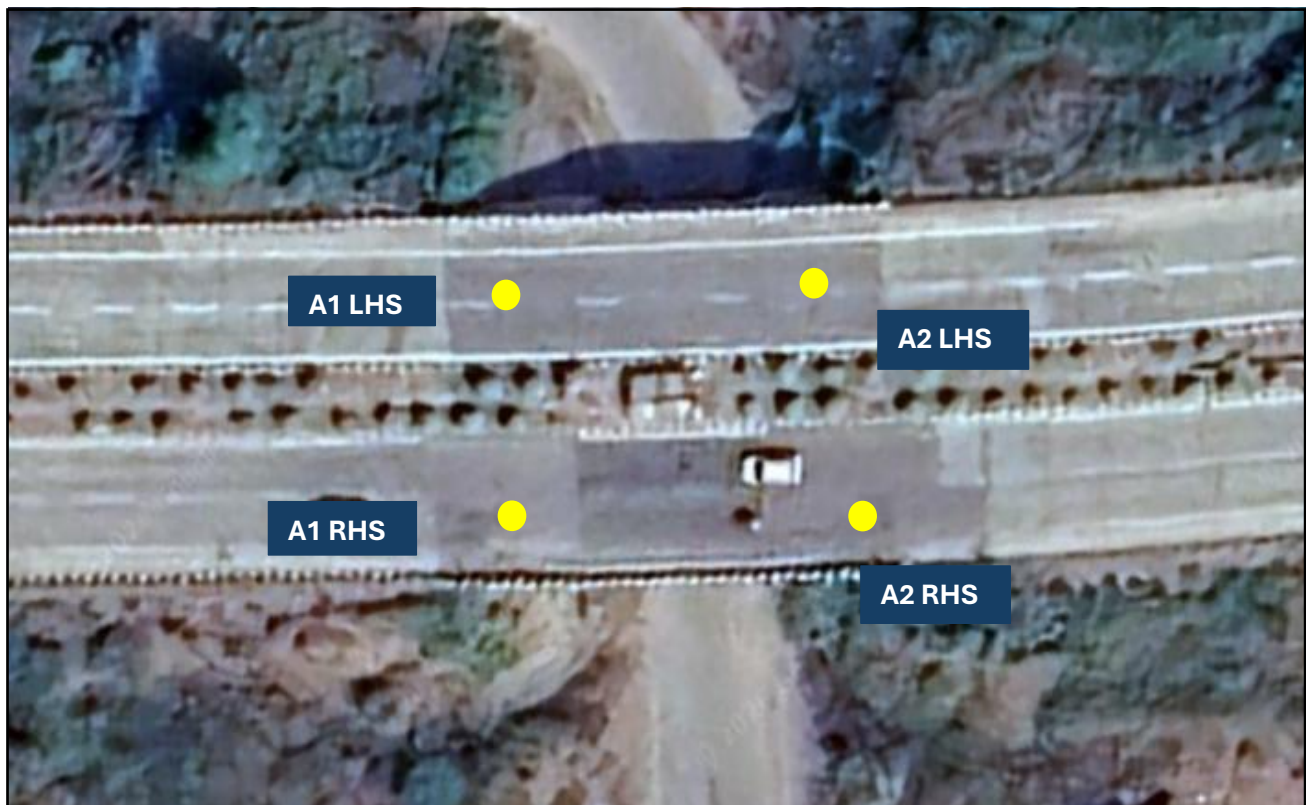


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

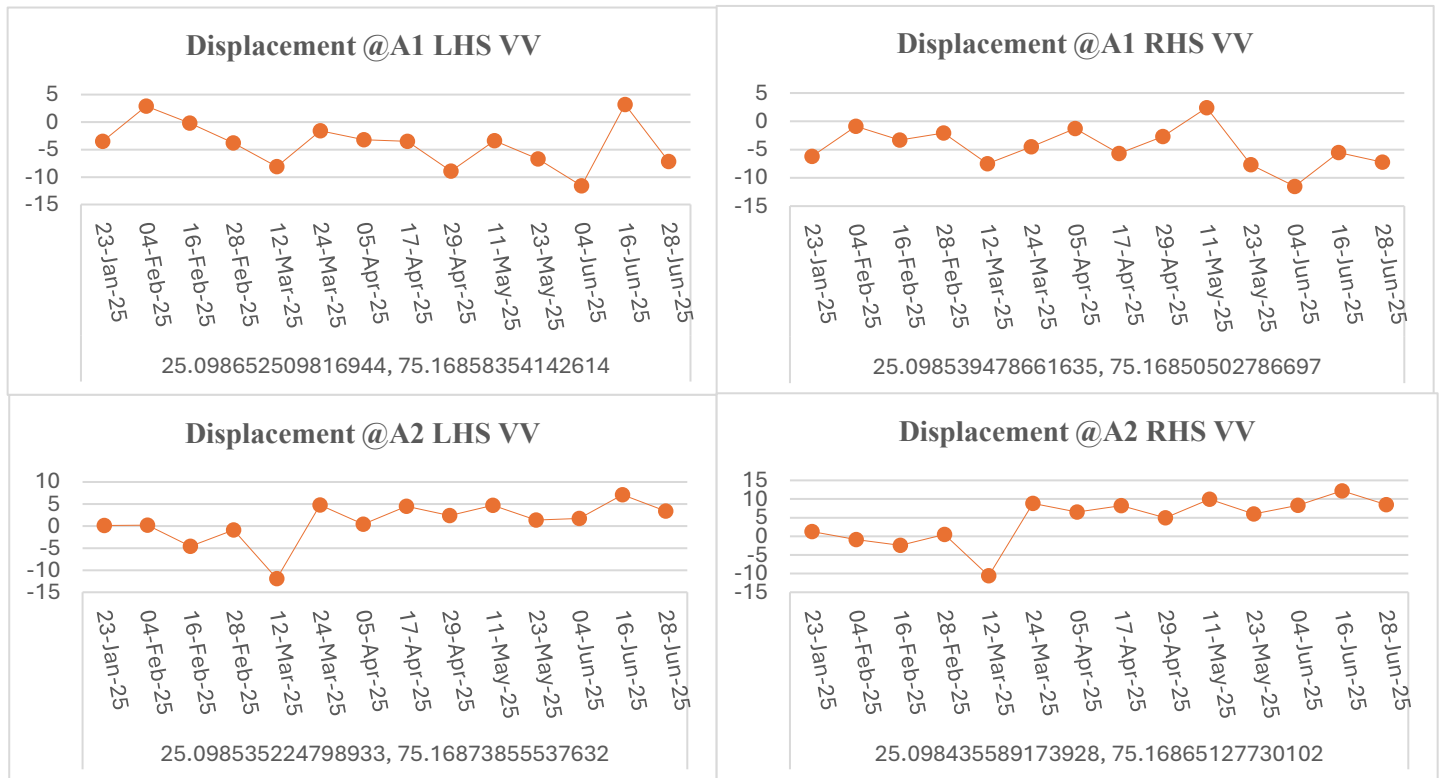
VUP- CH-983+400

Length – 30.67M

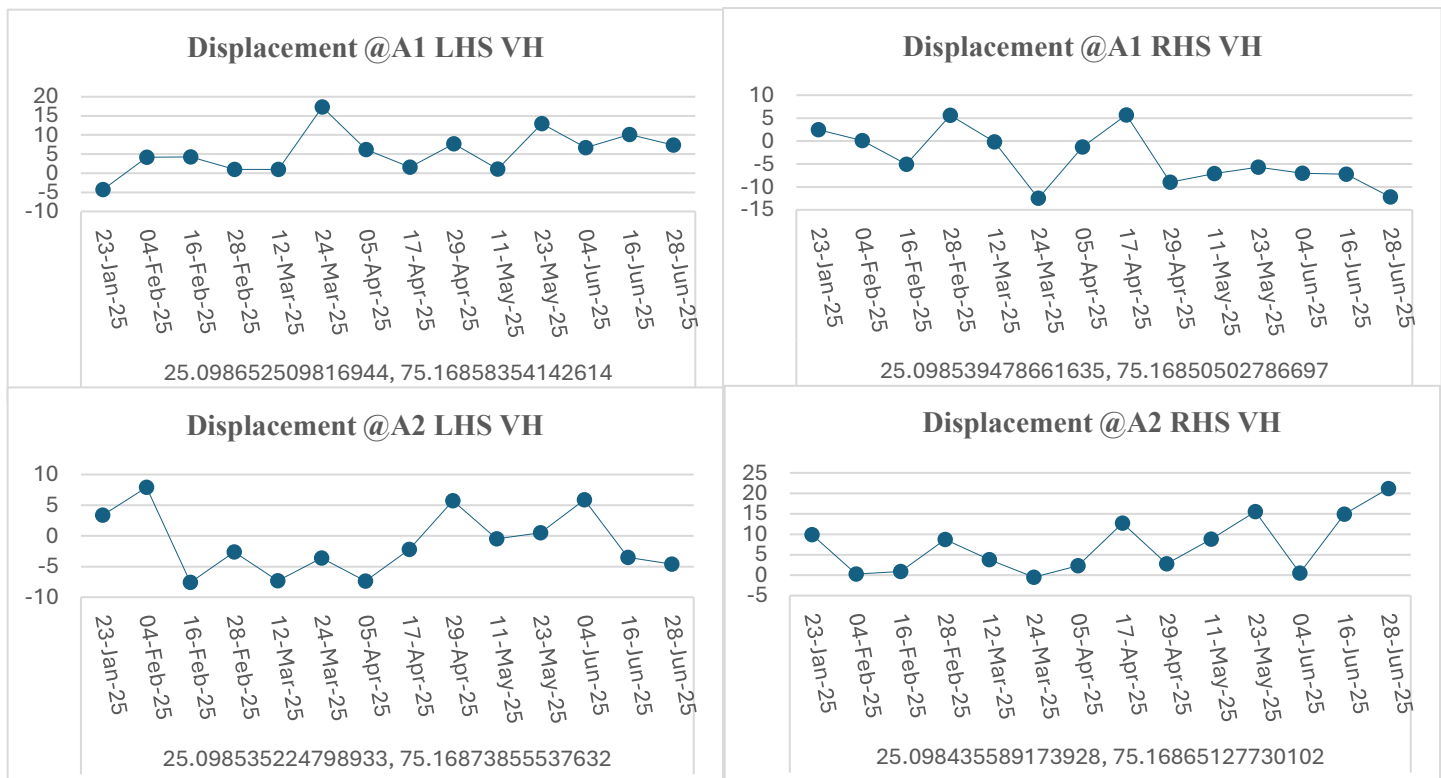


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



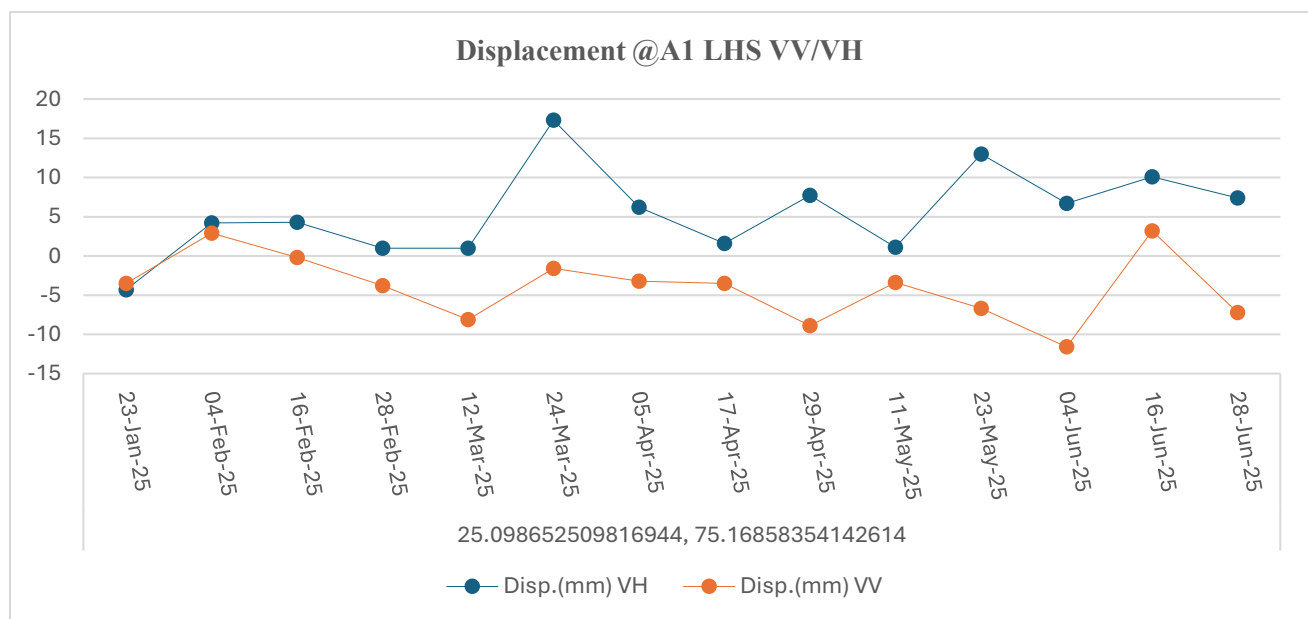
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.098652509816944, 75.16858354142614)



Key Observations:

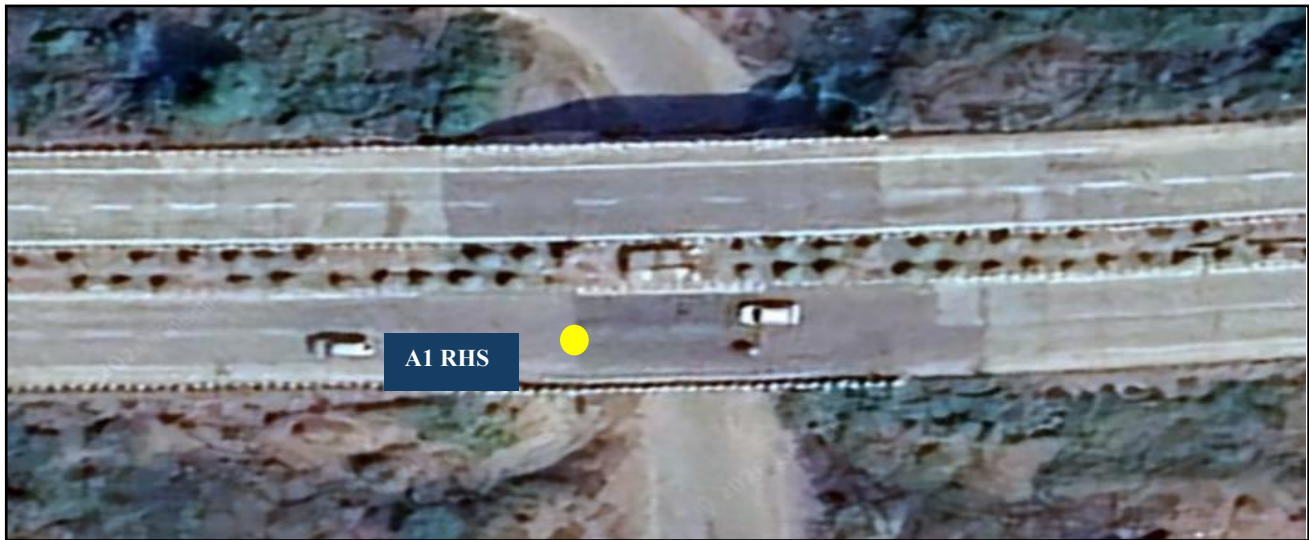
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits predominantly settlement-dominated behaviour with occasional minor uplift.
- **Horizontal Displacement (VH)** at A1 LHS shows moderate-to-high variability with repeated peaks, indicating cyclic lateral response.

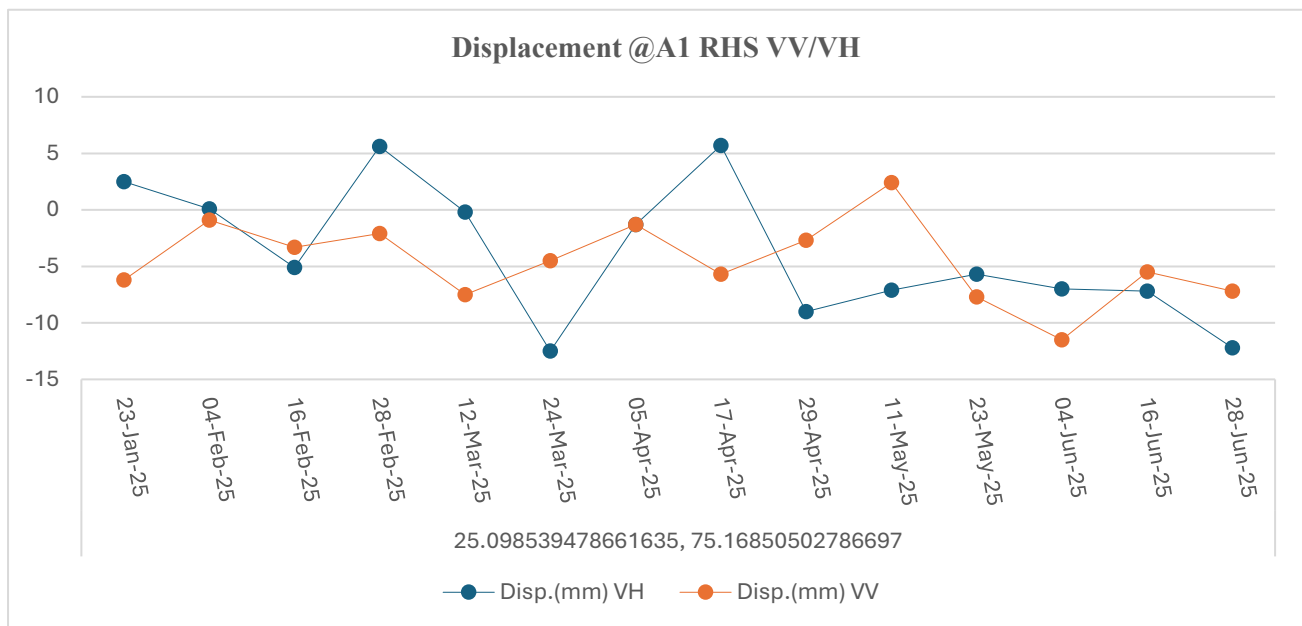
Critical Events & Anomalies:

- **January-February 2025:** VH starts near -5 mm and rises to about +4 mm, while VV briefly reaches +2 mm before transitioning into settlement, indicating initial stabilization followed by onset of vertical compression.
- **June 2025:** VH stabilizes between +7 mm and +10 mm, whereas VV shows a brief recovery to +3 mm before dropping again to -8 mm, indicating temporary vertical rebound followed by renewed settlement toward the end of monitoring.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.098539478661635, 75.16850502786697)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS shows pronounced oscillatory behaviour with sharp reversals, indicating an unstable lateral response compared to LHS.
- **Vertical displacement (VV)** Vertical displacement is predominantly settlement-controlled, with minor short-term uplift. VV values vary roughly between -12 mm and +2 mm.

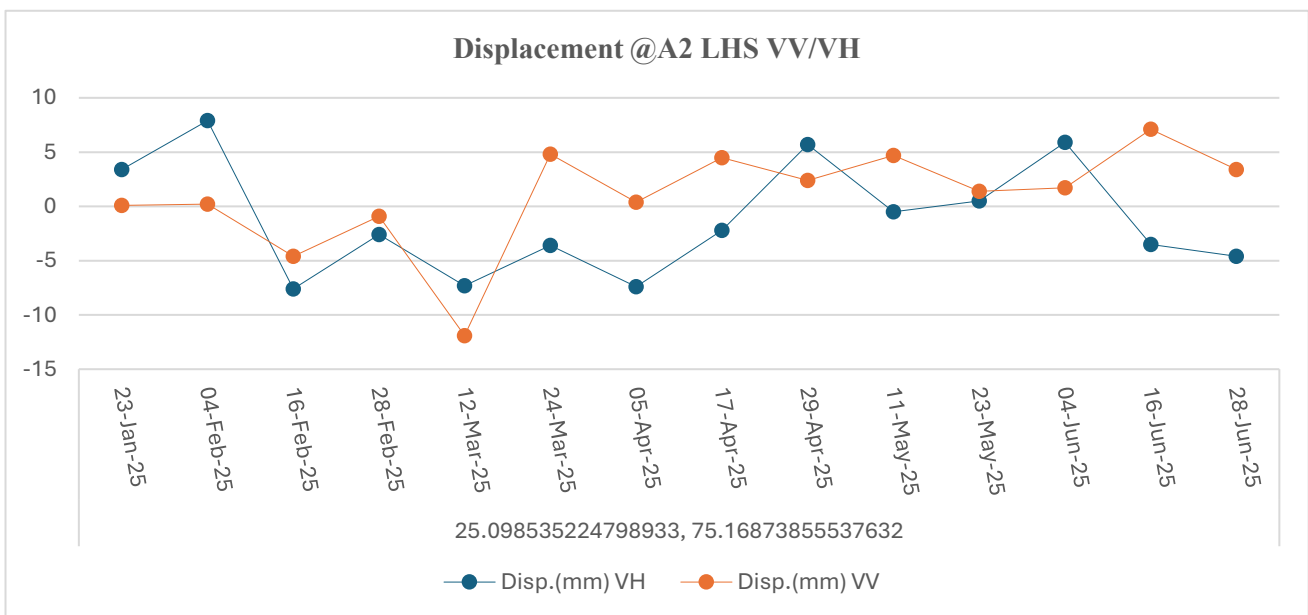
Critical Events & Anomalies:

- **January-February 2025:** VH begins near +2 mm and quickly transitions to -5 mm, while VV moves from -6 mm to -2 mm, indicating early lateral instability accompanied by vertical settlement.
- **April-May 2025:** VH partially recovers but remains negative (between -9 mm and -6 mm), whereas VV fluctuates and briefly reaches +2 mm in mid-May
- **June 2025:** Both VH and VV trend downward again, ending near -12 mm (VH) and -7 mm (VV), suggesting renewed settlement and lateral degradation toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.098535224798933, 75.16873855537632)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS shows moderate variability with frequent directional reversals, indicating a cyclic lateral response.
- **Vertical Displacement (VV)** Vertical displacement exhibits mixed uplift and settlement behaviour, with a wider spread compared to VH.

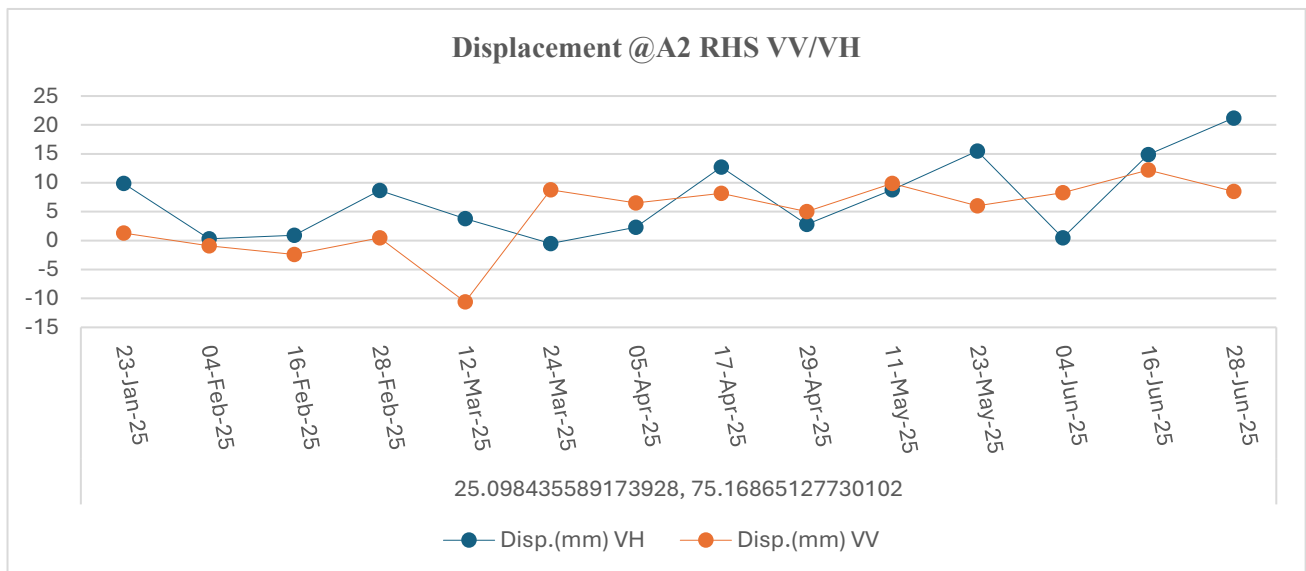
Critical Events & Anomalies:

- **January-February 2025:** VH starts positive (around +3 to +7 mm), while VV remains close to 0 mm, indicating initial lateral movement with negligible vertical response.
- **June 2025:** VH trends back to -4 to -5 mm, while VV reduces slightly, indicating renewed lateral softening with partial loss of vertical uplift.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.098435589173928, 75.16865127730102)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** displacement exhibits alternating settlement and uplift behaviour with moderate amplitude.
- **Horizontal (VH) displacement** at A2 RHS shows pronounced variability with an overall increasing trend toward the later monitoring period.

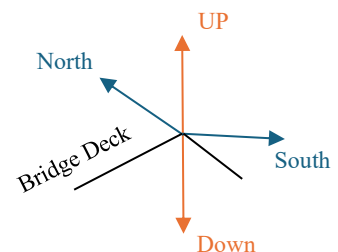
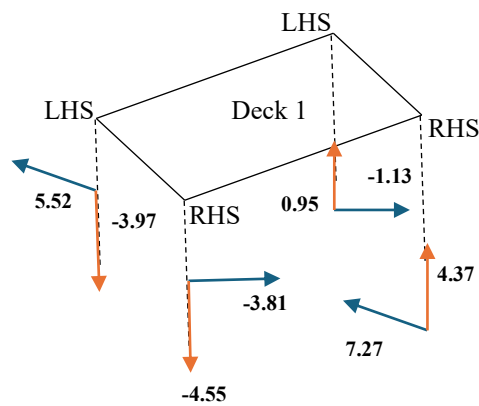
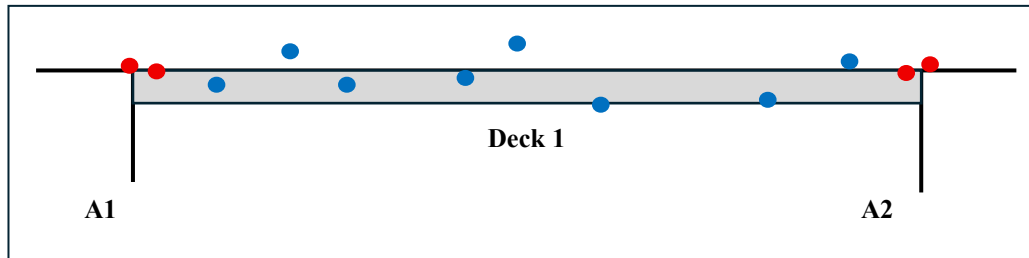
Critical Events & Anomalies:

- **January-February 2025:** VH starts at a relatively high positive value (+10 mm) while VV remains close to 0 mm, indicating initial dominant lateral displacement with minimal vertical involvement.
- **May 2025:** Both VH and VV increase simultaneously, with VH reaching +15 mm and VV stabilizing around +6 to +9 mm, indicating coupled horizontal and vertical deformation.
- **June 2025:** VH reaches its maximum value (+21 mm) by late June, while VV remains positive (+8–13 mm), indicating dominant lateral displacement with sustained vertical uplift toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS supports. The central deck region shows minimal uplift (approximately +0.95 mm), indicating near-neutral vertical behaviour at mid-span. The coexistence of uplift on the RHS and settlement on the LHS confirms vertical rotation of the deck slab rather than uniform bending.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability across the deck width, reflecting torsional deformation behaviour. The opposing directions and uneven magnitudes of horizontal displacement confirm torsional rotation about a longitudinal axis, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-influenced deformation response, characterized by asymmetric vertical uplift and settlement combined with differential horizontal displacement. Overall, the observed behaviour is moderate but structurally relevant, warranting continued monitoring and correlation with loading conditions and boundary restraints.